GraphTheorySymbol: A set of LaTeX commands to have symbols designed for Graph Theory

Laurent Frédéric Bernard François Lyaudet
Website:https://lyaudet.eu/laurent/
Email:Laurent.Lyaudet@gmail.com

v1.4.0 2026/09/12

Abstract

This documentation is for LaTeX package GraphTheorySymbol. It provides symbols used in graph theory. The adjacency and incidence symbols, and the meta-logical operators symbols were created by the author of this package.

Contents

1 Introduction

This documentation is for LaTeX package GraphTheorySymbol. It provides symbols used in graph theory. The adjacency and incidence symbols, and the meta-logical operators symbols were created by the author of this package.

This package is licenced under Lesser GNU General Public License version 3 or later (LGPLv3+). It means you are free do anything with it, but if you do improve the code of the symbols that we provide, please do share these improvements in a similar way. There is no restriction on other graph theory symbols/code, even if it would be a good fit for inclusion in this package. There is also no need to share in a similar way what you build with this package. The text of the licenses is available at:
https://www.gnu.org/licenses/.

The git repository of this source code is also available at:
https://github.com/LLyaudet/GraphTheorySymbol/.
Clone it with this URL:
git@github.com:LLyaudet/GraphTheorySymbol.git.

This package is available on CTAN:
https://ctan.org/pkg/graph-theory-symbol.

This is our first LaTeX package. It may be that our macros are quite slow and that a document making extensive use of these macros will be slow to compile. If you know how to produce better LaTeX code, please submit a pull request on GitHub.

2 Usage

2.1 Memo or for the impatient

Add in your preamble: `\usepackage{graph-theory-symbol}`.

For clarity, below we only show long macro names and short macro names; see Detailed usage section for intermediate length macro names.

Long name	Short name	Result	In context
`\GTSadjacency/`	`\GTSa/`		u v
`\GTSadjacencyRelation/`	`\GTSaR/`		u v
`\GTSunadjacency/`	`\GTSu/`		u v
`\GTSunadjacencyRelation/`	`\GTSuR/`		u v
`\GTSdirectedAdjacencyUp/`	`\GTSdAU/`		u v
`\GTSdirectedAdjacencyUpRelation/`	`\GTSdAUR/`		u v
`\GTSnegatedDirectedAdjacencyUp/`	`\GTSnDAU/`		u v
`\GTSnegatedDirectedAdjacencyUpRelation/`	`\GTSnDAUR/`		u v
`\GTSdirectedAdjacencyDown/`	`\GTSdAD/`		u v
`\GTSdirectedAdjacencyDownRelation/`	`\GTSdADR/`		u v
`\GTSnegatedDirectedAdjacencyDown/`	`\GTSnDAD/`		u v
`\GTSnegatedDirectedAdjacencyDownRelation/`	`\GTSnDADR/`		u v
`\GTSincidence/`	`\GTSi/`		v e
`\GTSincidenceRelation/`	`\GTSiR/`		v e
`\GTSnegatedIncidence/`	`\GTSnI/`		v e
`\GTSnegatedIncidenceRelation/`	`\GTSnIR/`		v e
`\GTSoutIncidence/`	`\GTSoI/`		v e
`\GTSoutIncidenceRelation/`	`\GTSoIR/`		v e
`\GTSnegatedOutIncidence/`	`\GTSnOI/`		v e
`\GTSnegatedOutIncidenceRelation/`	`\GTSnOIR/`		v e
`\GTSinIncidence/`	`\GTSiI/`		v e
`\GTSinIncidenceRelation/`	`\GTSiIR/`		v e
`\GTSnegatedInIncidence/`	`\GTSnII/`		v e
`\GTSnegatedInIncidenceRelation/`	`\GTSnIIR/`		v e

Long name	Short name	Result	In context
`\GTSadjacencyMonochrome/`	`\GTSaM/`		u v
`\GTSadjacencyMonochromeRelation/`	`\GTSaMR/`		u v
`\GTSunadjacencyMonochrome/`	`\GTSuM/`		u v
`\GTSunadjacencyMonochromeRelation/`	`\GTSuMR/`		u v
`\GTSdirectedAdjacencyUpMonochrome/`	`\GTSdAUM/`		u v
`\GTSdirectedAdjacencyUpMonochromeRelation/`	`\GTSdAUMR/`		u v
`\GTSnegatedDirectedAdjacencyUpMonochrome/`	`\GTSnDAUM/`		u v
`\GTSnegatedDirectedAdjacencyUpMonochromeRelation/`	`\GTSnDAUMR/`		u v
`\GTSdirectedAdjacencyDownMonochrome/`	`\GTSdADM/`		u v
`\GTSdirectedAdjacencyDownMonochromeRelation/`	`\GTSdADMR/`		u v
`\GTSnegatedDirectedAdjacencyDownMonochrome/`	`\GTSnDADM/`		u v
`\GTSnegatedDirectedAdjacencyDownMonochromeRelation/`	`\GTSnDADMR/`		u v
`\GTSincidenceMonochrome/`	`\GTSiM/`		v e
`\GTSincidenceMonochromeRelation/`	`\GTSiMR/`		v e
`\GTSnegatedIncidenceMonochrome/`	`\GTSnIM/`		v e
`\GTSnegatedIncidenceMonochromeRelation/`	`\GTSnIMR/`		v e
`\GTSoutIncidenceMonochrome/`	`\GTSoIM/`		v e
`\GTSoutIncidenceMonochromeRelation/`	`\GTSoIMR/`		v e
`\GTSnegatedOutIncidenceMonochrome/`	`\GTSnOIM/`		v e
`\GTSnegatedOutIncidenceMonochromeRelation/`	`\GTSnOIMR/`		v e
`\GTSinIncidenceMonochrome/`	`\GTSiIM/`		v e
`\GTSinIncidenceMonochromeRelation/`	`\GTSiIMR/`		v e
`\GTSnegatedInIncidenceMonochrome/`	`\GTSnIIM/`		v e
`\GTSnegatedInIncidenceMonochromeRelation/`	`\GTSnIIMR/`		v e

\GTSsetMainColor{blue}

Long name	Short name	Result	In context
\GTSadjacency/	\GTSa/		u v
\GTSadjacencyRelation/	\GTSaR/		u v
\GTSunadjacency/	\GTSu/		u v
\GTSunadjacencyRelation/	\GTSuR/		u v
\GTSdirectedAdjacencyUp/	\GTSdAU/		u v
\GTSdirectedAdjacencyUpRelation/	\GTSdAUR/		u v
\GTSnegatedDirectedAdjacencyUp/	\GTSnDAU/		u v
\GTSnegatedDirectedAdjacencyUpRelation/	\GTSnDAUR/		u v
\GTSdirectedAdjacencyDown/	\GTSdAD/		u v
\GTSdirectedAdjacencyDownRelation/	\GTSdADR/		u v
\GTSnegatedDirectedAdjacencyDown/	\GTSnDAD/		u v
\GTSnegatedDirectedAdjacencyDownRelation/	\GTSnDADR/		u v
\GTSincidence/	\GTSi/		v e
\GTSincidenceRelation/	\GTSiR/		v e
\GTSnegatedIncidence/	\GTSnI/		v e
\GTSnegatedIncidenceRelation/	\GTSnIR/		v e
\GTSoutIncidence/	\GTSoI/		v e
\GTSoutIncidenceRelation/	\GTSoIR/		v e
\GTSnegatedOutIncidence/	\GTSnOI/		v e
\GTSnegatedOutIncidenceRelation/	\GTSnOIR/		v e
\GTSinIncidence/	\GTSiI/		v e
\GTSinIncidenceRelation/	\GTSiIR/		v e
\GTSnegatedInIncidence/	\GTSnII/		v e
\GTSnegatedInIncidenceRelation/	\GTSnIIR/		v e

Long name	Short name	Result	In context
\GTSmetaNot/	\GTSmN/		
\GTSmetaAnd/	\GTSmA/		
\GTSmetaOr/	\GTSmO/		

Implicit meta-and via concatenation, no space except around whole relations compositions, no parentheses via higher priority to meta-and, brackets when all possible adjacencies are specified between two vertices:

Tournament adjacency:

$\forall u, v, u$ [ ] v.

Tournament adjacency with maybe some colored edge also, who knows?:

$\forall u, v, u$ v.

Directed graph:

$\forall u, v, u$ [ ] v.

Oriented graph:

$\forall u, v, u$ [ ] v.

Prefix notation:

$\exists u, v, u$ [ ] v.

$\exists v, e, v$ [ ] e.

2.2 Detailed usage

First you need to add in your preamble: `\usepackage{graph-theory-symbol}`.

All symbols macros were designed with ease of use in mind: they are called with a trailing slash, they can be called in text mode or in math mode, a variant with relation spacing for math mode exists with suffix "relation".

`\GTS@newcommand` This macro will enforce that the other macros are called with a trailing slash, avoiding problems with spaces after macro call.
Arguments:
`{`⟨*commandname*⟩`}` The name of the command to create.
`{`⟨*commandcode*⟩`}` The code of the command to create.

`\GTS@newprotectedcommand` This macro will enforce that the other macros are called with a trailing slash, avoiding problems with spaces after macro call.
Arguments:
`{`⟨*commandname*⟩`}` The name of the command to create.
`{`⟨*commandcode*⟩`}` The code of the command to create.

`\GTSsetMainColor` This macro will set the main color used in GraphTheorySymbol. This color has initial value "black".
Argument:
`{`⟨*color*⟩`}` The color that must be used by GraphTheorySymbol afterward.

`\GTSsetSecondColor` This macro will set the second color used in GraphTheorySymbol. This color has initial value "gray".
Argument:
`{`⟨*color*⟩`}` The color that must be used by GraphTheorySymbol afterward.

`\GTS@a` This macro will draw any of the adjacency symbols if the correct arguments are given. It is not expected to be used directly. See the macros derived from it and `\GTS@b`: `\GTSxadjacency`, `\GTSxadj`, and `\GTSxa`.
Arguments:
`{`⟨*mainColor*⟩`}` If non-empty, this color will be used as main color.
`{`⟨*secondColor*⟩`}` If non-empty, this color will be used as second color.
`{`⟨*drawSlash*⟩`}` Non-empty implies that a negation will be drawn.
`{`⟨*drawUpArrow*⟩`}` Non-empty implies that an up-arrow will be drawn.
`{`⟨*drawDownArrow*⟩`}` Non-empty implies that a down-arrow will be drawn.

`\GTS@b` This macro builds on top of `\GTS@a` to handle mathematical relation spacing if sixth argument is not empty. It is not expected to be used directly. See the macros derived from it and `\GTS@a`: `\GTSxadjacency`, `\GTSxadj`, and `\GTSxa`.
Arguments:
`{`⟨*mainColor*⟩`}` If non-empty, this color will be used as main color.
`{`⟨*secondColor*⟩`}` If non-empty, this color will be used as second color.
`{`⟨*drawSlash*⟩`}` Non-empty implies that a negation will be drawn.
`{`⟨*drawUpArrow*⟩`}` Non-empty implies that an up-arrow will be drawn.
`{`⟨*drawDownArrow*⟩`}` Non-empty implies that a down-arrow will be drawn.
`{`⟨*relation*⟩`}` Non-empty implies that relation spacing will be added.

`\GTSxa`
`\GTSxadj`
`\GTSxadjacency`
Versatile macros for adjacency that can be given options.
Argument:
`[`⟨*optionsList*⟩`]` Sets the following options if not empty:
mainColor=color1 or mainC=color1 or mColor=color1 or mC=color1,
secondColor=color2 or secondC=color2 or sColor=color2 or sC=color2,
monochrome or mono or mon or m (equivalent to mC=black,sC=black),
negated or not or n,
up or u,
down or d,
relation or rel or r.

`\GTSadjacency`
`\GTSadj`
`\GTSa`
This macro will draw an adjacency symbol. A first drawing didn't include the small dots in the middle of the two circles corresponding to the graph vertices. But it was too similar to " spoon" and multimap symbols, and could be confused with a graph with only two vertices and one edge. Hence, these dots were added and they have the nice property

to remind of the classical textual notation of adjacency relation as Adj(.,.).
`A.\GTSadjacency/A.`
or `A.\GTSadj/A.`
or `A.\GTSa/A.`
will yield A.A.
`u \GTSadjacency/ v`
or `u \GTSadj/ v`
or `u \GTSa/ v`
will yield u v.
`$u \GTSadjacency/ v$`
or `$u \GTSadj/ v$`
or `$u \GTSa/ v$`
will yield $u v$.

`\GTSadjacencyRelation`
`\GTSadjRel`
`\GTSaR`

This macro will draw an adjacency symbol.
`$u\GTSadjacencyRelation/v$`
or `$u\GTSadjRel/v$`
or `$u\GTSaR/v$`
will yield $u \; v$.

`\GTSunadjacency`
`\GTSunadj`
`\GTSu`

This macro will draw an unadjacency symbol.
`A.\GTSunadjacency/A.`
or `A.\GTSunadj/A.`
or `A.\GTSu/A.`
will yield A.A.
`u \GTSunadjacency/ v`
or `u \GTSunadj/ v`
or `u \GTSu/ v`
will yield u v.
`$u \GTSunadjacency/ v$`
or `$u \GTSunadj/ v$`
or `$u \GTSu/ v$`
will yield $u v$.

`\GTSunadjacencyRelation`
`\GTSunadjRel`
`\GTSuR`

This macro will draw an unadjacency symbol.
`$u\GTSunadjacencyRelation/v$`
or `$u\GTSunadjRel/v$`
or `$u\GTSuR/v$`
will yield $u \; v$.

`\GTSdirectedAdjacencyUp`
`\GTSdirAdjUp`
`\GTSdAU`

This macro will draw a directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.
`A.\GTSdirectedAdjacencyUp/A.`
or `A.\GTSdirAdjUp/A.`
or `A.\GTSdAU/A.`
will yield A.A.
`u \GTSdirectedAdjacencyUp/ v`
or `u \GTSdirAdjUp/ v`
or `u \GTSdAU/ v`
will yield u v.
`$u \GTSdirectedAdjacencyUp/ v$`
or `$u \GTSdirAdjUp/ v$`
or `$u \GTSdAU/ v$`
will yield $u v$.

`\GTSdirectedAdjacencyUp-Relation`
`\GTSdirAdjUpRel`
`\GTSdAUR`

This macro will draw a directed adjacency upward symbol with relation spacing in math mode.
`$u\GTSdirectedAdjacencyUpRelation/v$`
or `$u\GTSdirAdjUpRel/v$`
or `$u\GTSdAUR/v$`
will yield $u \; v$.

`\GTSnegatedDirectedAdjacencyUp`
`\GTSnegDirAdjUp`
`\GTSnDAU`

This macro will draw a negated directed adjacency upward symbol.
`A.\GTSnegatedDirectedAdjacencyUp/A.`
or `A.\GTSnegDirAdjUp/A.`
or `A.\GTSnDAU/A.`
will yield A.$\not\uparrow$A.
`u \GTSnegatedDirectedAdjacencyUp/ v`
or `u \GTSnegDirAdjUp/ v`
or `u \GTSnDAU/ v`
will yield u $\not\uparrow$ v.
`$u \GTSnegatedDirectedAdjacencyUp/ v$`
or `$u \GTSnegDirAdjUp/ v$`
or `$u \GTSnDAU/ v$`
will yield $u{\not\uparrow}v$.

`\GTSnegatedDirectedAdjacencyUpRelation`
`\GTSnegDirAdjUpRel`
`\GTSnDAUR`

This macro will draw a negated directed adjacency upward symbol with relation spacing in math mode.
`$u\GTSnegatedDirectedAdjacencyUpRelation/v$`
or `$u\GTSnegDirAdjUpRel/v$`
or `$u\GTSnDAUR/v$`
will yield $u \not\uparrow v$.

`\GTSdirectedAdjacencyDown`
`\GTSdirAdjDown`
`\GTSdAD`

This macro will draw a directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.
`A.\GTSdirectedAdjacencyDown/A.`
or `A.\GTSdirAdjDown/A.`
or `A.\GTSnDAD/A.`
will yield A.$\downarrow$A.
`u \GTSdirectedAdjacencyDown/ v`
or `u \GTSdirAdjDown/ v`
or `u \GTSnDAD/ v`
will yield u $\downarrow$ v.
`$u \GTSdirectedAdjacencyDown/ v$`
or `$u \GTSdirAdjDown/ v$`
or `$u \GTSnDAD/ v$`
will yield $u{\downarrow}v$.

`\GTSdirectedAdjacencyDownRelation`
`\GTSdirAdjDownRel`
`\GTSdADR`

This macro will draw a directed adjacency downward symbol with relation spacing in math mode.
`$u\GTSdirectedAdjacencyDownRelation/v$`
or `$u\GTSdirAdjDownRel/v$`
or `$u\GTSdADR/v$`
will yield $u \downarrow v$.

`\GTSnegatedDirectedAdjacencyDown`
`\GTSnegDirAdjDown`
`\GTSnDAD`

This macro will draw a negated directed adjacency downward symbol.
`A.\GTSnegatedDirectedAdjacencyDown/A.`
or `A.\GTSnegDirAdjDown/A.`
or `A.\GTSnDAD/A.`
will yield A.$\not\downarrow$A.
`u \GTSnegatedDirectedAdjacencyDown/ v`
or `u \GTSnegDirAdjDown/ v`
or `u \GTSnDAD/ v`
will yield u $\not\downarrow$ v.
`$u \GTSnegatedDirectedAdjacencyDown/ v$`
or `$u \GTSnegDirAdjDown/ v$`
or `$u \GTSnDAD/ v$`
will yield $u{\not\downarrow}v$.

`\GTSnegatedDirectedAdjacencyDownRelation`
`\GTSnegDirAdjDownRel`
`\GTSnDADR`

This macro will draw a negated directed adjacency downward symbol with relation spacing in math mode.
`$u\GTSnegatedDirectedAdjacencyDownRelation/v$`
or `$u\GTSnegDirAdjDownRel/v$`
or `$u\GTSnDADR/v$`
will yield $u \not\downarrow v$.

`\GTS@i` This macro will draw any of the incidence symbols if the correct arguments are given. It is not expected to be used directly. See the macros derived from it and `\GTS@j`: `\GTSxincidence`, `\GTSxinc`, and `\GTSxi`. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.
Arguments:
`{`⟨*mainColor*⟩`}` If non-empty, this color will be used as main color.
`{`⟨*secondColor*⟩`}` If non-empty, this color will be used as second color.
`{`⟨*drawSlash*⟩`}` Non-empty implies that a negation will be drawn.
`{`⟨*drawUpArrow*⟩`}` Non-empty implies that an up-arrow will be drawn.
`{`⟨*drawDownArrow*⟩`}` Non-empty implies that a down-arrow will be drawn.

`\GTS@j` This macro builds on top of `\GTS@i` to handle mathematical relation spacing if sixth argument is not empty. It is not expected to be used directly. See the macros derived from it and `\GTS@i`: `\GTSxincidence`, `\GTSxinc`, and `\GTSxi`.
Arguments:
`{`⟨*mainColor*⟩`}` If non-empty, this color will be used as main color.
`{`⟨*secondColor*⟩`}` If non-empty, this color will be used as second color.
`{`⟨*drawSlash*⟩`}` Non-empty implies that a negation will be drawn.
`{`⟨*drawUpArrow*⟩`}` Non-empty implies that an up-arrow will be drawn.
`{`⟨*drawDownArrow*⟩`}` Non-empty implies that a down-arrow will be drawn.
`{`⟨*relation*⟩`}` Non-empty implies that relation spacing will be added.

`\GTSxi`
`\GTSxinc`
`\GTSxincidence`
Versatile macros for incidence that can be given options.
Argument:
`[`⟨*optionsList*⟩`]` Sets the following options if not empty:
mainColor=color1 or mainC=color1 or mColor=color1 or mC=color1,
secondColor=color2 or secondC=color2 or sColor=color2 or sC=color2,
monochrome or mono or mon or m (equivalent to mC=black,sC=black),
negated or not or n,
in or i,
out or o,
relation or rel or r.

`\GTSincidence`
`\GTSinc`
`\GTSi`
This macro will draw an incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.
`A.\GTSincidence/A.`
or `A.\GTSinc/A.`
or `A.\GTSi/A.`
will yield A.A.
`v \GTSincidence/ e`
or `v \GTSinc/ e`
or `v \GTSi/ e`
will yield v e.
`$v \GTSincidence/ e$`
or `$v \GTSinc/ e$`
or `$v \GTSi/ e$`
will yield ve.

`\GTSincidenceRelation`
`\GTSincRel`
`\GTSiR`
This macro will draw an incidence symbol with relation spacing in math mode.
`$v\GTSincidenceRelation/e$`
or `$v\GTSincRel/e$`
or `$v\GTSiR/e$`
will yield $v \; e$.

`\GTSnegatedIncidence`
`\GTSnegInc`
`\GTSnI`
This macro will draw a negated incidence symbol.
`A.\GTSnegatedIncidence/A.`
or `A.\GTSnegInc/A.`
or `A.\GTSnI/A.`
will yield A.A.
`v \GTSnegatedIncidence/ e`
or `v \GTSnegInc/ e`
or `v \GTSnI/ e`
will yield v e.

`$v \GTSnegatedIncidence/ e$`
or `$v \GTSnegInc/ e$`
or `$v \GTSnI/ e$`
will yield v e.

`\GTSnegatedIncidence-Relation` `\GTSnegIncRel` `\GTSnIR`

This macro will draw a negated incidence symbol with relation spacing in math mode.
`$v\GTSnegatedIncidenceRelation/e$`
or `$v\GTSnegIncRel/e$`
or `$v\GTSnIR/e$`
will yield v e.

`\GTSoutIncidence` `\GTSoutInc` `\GTSoI`

This macro will draw a directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.
`A.\GTSoutIncidence/A.`
or `A.\GTSoutInc/A.`
or `A.\GTSoI/A.`
will yield A.A.
`v \GTSoutIncidence/ e`
or `v \GTSoutInc/ e`
or `v \GTSoI/ e`
will yield v e.
`$v \GTSoutIncidence/ e$`
or `$v \GTSoutInc/ e$`
or `$v \GTSoI/ e$`
will yield v e.

`\GTSoutIncidenceRelation` `\GTSoutIncRel` `\GTSoIR`

This macro will draw a directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.
`$v\GTSoutIncidenceRelation/e$`
or `$v\GTSoutIncRel/e$`
or `$v\GTSoIR/e$`
will yield v e.

`\GTSnegatedOutIncidence` `\GTSnegOutInc` `\GTSnOI`

This macro will draw a negated directed incidence upward (out(ward) incidence) symbol.
`A.\GTSnegatedOutIncidence/A.`
or `A.\GTSnegOutInc/A.`
or `A.\GTSnOI/A.`
will yield A.A.
`v \GTSnegatedOutIncidence/ e`
or `v \GTSnegOutInc/ e`
or `v \GTSnOI/ e`
will yield v e.
`$v \GTSnegatedOutIncidence/ e$`
or `$v \GTSnegOutInc/ e$`
or `$v \GTSnOI/ e$`
will yield v e.

`\GTSnegatedOutIncidence-Relation` `\GTSnegOutIncRel` `\GTSnOIR`

This macro will draw a negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.
`$v\GTSnegatedOutIncidenceRelation/e$`
or `$v\GTSnegOutIncRel/e$`
or `$v\GTSnOIR/e$`
will yield v e.

`\GTSinIncidence` `\GTSinInc` `\GTSiI`

This macro will draw a directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.
`A.\GTSinIncidence/A.`
or `A.\GTSinInc/A.`
or `A.\GTSiI/A.`
will yield A.A.
`v \GTSinIncidence/ e`
or `v \GTSinInc/ e`

or v \GTSiI/ e
will yield v e.
$v \GTSinIncidence/ e$
or $v \GTSinInc/ e$
or $v \GTSiI/ e$
will yield ve.

`\GTSinIncidenceRelation` `\GTSinIncRel` `\GTSiIR` This macro will draw a directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.
$v\GTSinIncidenceRelation/e$
or $v\GTSinIncRel/e$
or $v\GTSiIR/e$
will yield $v\ e$.

`\GTSnegatedInIncidence` `\GTSnegInInc` `\GTSnII` This macro will draw a negated directed incidence downward (in(ward) incidence) symbol.
A.\GTSnegatedInIncidence/A.
or A.\GTSnegInInc/A.
or A.\GTSnII/A.
will yield A.A.
v \GTSnegatedInIncidence/ e
or v \GTSnegInInc/ e
or v \GTSnII/ e
will yield v e.
$v \GTSnegatedInIncidence/ e$
or $v \GTSnegInInc/ e$
or $v \GTSnII/ e$
will yield ve.

`\GTSnegatedInIncidenceRelation` `\GTSnegInIncRel` `\GTSnIIR` This macro will draw a negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.
$v\GTSnegatedInIncidenceRelation/e$
or $v\GTSnegInIncRel/e$
or $v\GTSnIIR/e$
will yield $v\ e$.

`\GTSadjacencyMonochrome` `\GTSadjMono` `\GTSadjMon` `\GTSaM` This macro will draw a monochrome adjacency symbol.
A.\GTSadjacencyMonochrome/A.
or A.\GTSadjMono/A.
or A.\GTSadjMon/A.
or A.\GTSaM/A.
will yield A.A.
u \GTSadjacencyMonochrome/ v
or u \GTSadjMono/ v
or u \GTSadjMon/ v
or u \GTSaM/ v
will yield u v.
$u \GTSadjacencyMonochrome/ v$
or $u \GTSadjMono/ v$
or $u \GTSadjMon/ v$
or $u \GTSaM/ v$
will yield uv.

`\GTSadjacencyMonochromeRelation` `\GTSadjMonoRel` `\GTSadjMonRel` `\GTSaMR` This macro will draw a monochrome adjacency symbol with relation spacing in math mode.
$u\GTSadjacencyMonochromeRelation/v$
or $u\GTSadjMonoRel/v$
or $u\GTSadjMonRel/v$
or $u\GTSaMR/v$
will yield $u\ v$.

`\GTSunadjacencyMonochrome` `\GTSunadjMono` `\GTSunadjMon` `\GTSuM` This macro will draw a monochrome unadjacency symbol.
A.\GTSunadjacencyMonochrome/A.
or A.\GTSunadjMono/A.

or `A.\GTSunadjMon/A.`
or `A.\GTSuM/A.`
will yield A.A.
`u \GTSunadjacencyMonochrome/ v`
or `u \GTSunadjMono/ v`
or `u \GTSunadjMon/ v`
or `u \GTSuM/ v`
will yield u v.
`$u \GTSunadjacencyMonochrome/ v$`
or `$u \GTSunadjMono/ v$`
or `$u \GTSunadjMon/ v$`
or `$u \GTSuM/ v$`
will yield uv.

`\GTSunadjacencyMonochromeRelation`
`\GTSunadjMonoRel`
`\GTSunadjMonRel`
`\GTSuMR`

This macro will draw a monochrome unadjacency symbol with relation spacing in math mode.
`$u\GTSunadjacencyMonochromeRelation/v$`
or `$u\GTSunadjMonoRel/v$`
or `$u\GTSunadjMonRel/v$`
or `$u\GTSuMR/v$`
will yield $u \; v$.

`\GTSdirectedAdjacencyUpMonochrome`
`\GTSdirAdjUpMono`
`\GTSdirAdjUpMon`
`\GTSdAUM`

This macro will draw a monochrome directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.
`A.\GTSdirectedAdjacencyUpMonochrome/A.`
or `A.\GTSdirAdjUpMono/A.`
or `A.\GTSdirAdjUpMon/A.`
or `A.\GTSdAUM/A.`
will yield A.A.
`u \GTSdirectedAdjacencyUpMonochrome/ v`
or `u \GTSdirAdjUpMono/ v`
or `u \GTSdirAdjUpMon/ v`
or `u \GTSdAUM/ v`
will yield u v.
`$u \GTSdirectedAdjacencyUpMonochrome/ v$`
or `$u \GTSdirAdjUpMono/ v$`
or `$u \GTSdirAdjUpMon/ v$`
or `$u \GTSdAUM/ v$`
will yield uv.

`\GTSdirectedAdjacencyUpMonochromeRelation`
`\GTSdirAdjUpMonoRel`
`\GTSdirAdjUpMonRel`
`\GTSdAUMR`

This macro will draw a monochrome directed adjacency upward symbol with relation spacing in math mode.
`$u\GTSdirectedAdjacencyUpMonochromeRelation/v$`
or `$u\GTSdirAdjUpMonoRel/v$`
or `$u\GTSdirAdjUpMonRel/v$`
or `$u\GTSdAUMR/v$`
will yield $u \; v$.

`\GTSnegatedDirectedAdjacencyUpMonochrome`
`\GTSnegDirAdjUpMono`
`\GTSnegDirAdjUpMon`
`\GTSnDAUM`

This macro will draw a monochrome negated directed adjacency upward symbol.
`A.\GTSnegatedDirectedAdjacencyUpMonochrome/A.`
or `A.\GTSnegDirAdjUpMono/A.`
or `A.\GTSnegDirAdjUpMon/A.`
or `A.\GTSnDAUM/A.`
will yield A.A.
`u \GTSnegatedDirectedAdjacencyUpMonochrome/ v`
or `u \GTSnegDirAdjUpMono/ v`
or `u \GTSnegDirAdjUpMon/ v`
or `u \GTSnDAUM/ v`
will yield u v.
`$u \GTSnegatedDirectedAdjacencyUpMonochrome/ v$`
or `$u \GTSnegDirAdjUpMono/ v$`
or `$u \GTSnegDirAdjUpMon/ v$`

or `$u \GTSnDAUM/ v$`
will yield uv.

`\GTSnegatedDirectedAdjacencyUpMonochromeRelation` `\GTSnegDirAdjUpMonoRel` `\GTSnegDirAdjUpMonRel` `\GTSnDAUMR`

This macro will draw a monochrome negated directed adjacency upward symbol with relation spacing in math mode.
`$u\GTSnegatedDirectedAdjacencyUpMonochromeRelation/v$`
or `$u\GTSnegDirAdjUpMonoRel/v$`
or `$u\GTSnegDirAdjUpMonRel/v$`
or `$u\GTSnDAUMR/v$`
will yield $u \; v$.

`\GTSdirectedAdjacencyDownMonochrome` `\GTSdirAdjDownMono` `\GTSdirAdjDownMon` `\GTSdADM`

This macro will draw a monochrome directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.
`A.\GTSdirectedAdjacencyDownMonochrome/A.`
or `A.\GTSdirAdjDownMono/A.`
or `A.\GTSdirAdjDownMon/A.`
or `A.\GTSdADM/A.`
will yield A.A.
`u \GTSdirectedAdjacencyDownMonochrome/ v`
or `u \GTSdirAdjDownMono/ v`
or `u \GTSdirAdjDownMon/ v`
or `u \GTSdADM/ v`
will yield u v.
`$u \GTSdirectedAdjacencyDownMonochrome/ v$`
or `$u \GTSdirAdjDownMono/ v$`
or `$u \GTSdirAdjDownMon/ v$`
or `$u \GTSdADM/ v$`
will yield uv.

`\GTSdirectedAdjacencyDownMonochromeRelation` `\GTSdirAdjDownMonoRel` `\GTSdirAdjDownMonRel` `\GTSdADMR`

This macro will draw a monochrome directed adjacency downward symbol with relation spacing in math mode.
`$u\GTSdirectedAdjacencyDownMonochromeRelation/v$`
or `$u\GTSdirAdjDownMonoRel/v$`
or `$u\GTSdirAdjDownMonRel/v$`
or `$u\GTSdADMR/v$`
will yield $u \; v$.

`\GTSnegatedDirectedAdjacencyDownMonochrome` `\GTSnegDirAdjDownMono` `\GTSnegDirAdjDownMon` `\GTSnDADM`

This macro will draw a monochrome negated directed adjacency downward symbol.
`A.\GTSnegatedDirectedAdjacencyDownMonochrome/A.`
or `A.\GTSnegDirAdjDownMono/A.`
or `A.\GTSnegDirAdjDownMon/A.`
or `A.\GTSnDADM/A.`
will yield A.A.
`u \GTSnegatedDirectedAdjacencyDownMonochrome/ v`
or `u \GTSnegDirAdjDownMono/ v`
or `u \GTSnegDirAdjDownMon/ v`
or `u \GTSnDADM/ v`
will yield u v.
`$u \GTSnegatedDirectedAdjacencyDownMonochrome/ v$`
or `$u \GTSnegDirAdjDownMono/ v$`
or `$u \GTSnegDirAdjDownMon/ v$`
or `$u \GTSnDADM/ v$`
will yield uv.

`\GTSnegatedDirectedAdjacencyDownMonochromeRelation` `\GTSnegDirAdjDownMonoRel` `\GTSnegDirAdjDownMonRel` `\GTSnDADMR`

This macro will draw a monochrome negated directed adjacency downward symbol with relation spacing in math mode.
`$u\GTSnegatedDirectedAdjacencyDownMonochromeRelation/v$`
or `$u\GTSnegDirAdjDownMonoRel/v$`
or `$u\GTSnegDirAdjDownMonRel/v$`
or `$u\GTSnDADMR/v$`
will yield $u \; v$.

`\GTSincidenceMonochrome`
`\GTSincMono`
`\GTSincMon`
`\GTSiM`

This macro will draw a monochrome incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.
`A.\GTSincidenceMonochrome/A.`
or `A.\GTSincMono/A.`
or `A.\GTSincMon/A.`
or `A.\GTSiM/A.`
will yield A.⫰A.
`v \GTSincidenceMonochrome/ e`
or `v \GTSincMono/ e`
or `v \GTSincMon/ e`
or `v \GTSiM/ e`
will yield v ⫰ e.
`$v \GTSincidenceMonochrome/ e$`
or `$v \GTSincMono/ e$`
or `$v \GTSincMon/ e$`
or `$v \GTSiM/ e$`
will yield *v*⫰*e*.

`\GTSincidenceMonochrome-Relation`
`\GTSincMonoRel`
`\GTSincMonRel`
`\GTSiMR`

This macro will draw a monochrome incidence symbol with relation spacing in math mode.
`$v\GTSincidenceMonochromeRelation/e$`
or `$v\GTSincMonoRel/e$`
or `$v\GTSincMonRel/e$`
or `$v\GTSiMR/e$`
will yield *v* ⫰ *e*.

`\GTSnegatedIncidence-Monochrome`
`\GTSnegIncMono`
`\GTSnegIncMon`
`\GTSnIM`

This macro will draw a monochrome negated incidence symbol.
`A.\GTSnegatedIncidenceMonochrome/A.`
or `A.\GTSnegIncMono/A.`
or `A.\GTSnegIncMon/A.`
or `A.\GTSnIM/A.`
will yield A.⫰̸A.
`v \GTSnegatedIncidenceMonochrome/ e`
or `v \GTSnegIncMono/ e`
or `v \GTSnegIncMon/ e`
or `v \GTSnIM/ e`
will yield v ⫰̸ e.
`$v \GTSnegatedIncidenceMonochrome/ e$`
or `$v \GTSnegIncMono/ e$`
or `$v \GTSnegIncMon/ e$`
or `$v \GTSnIM/ e$`
will yield *v*⫰̸*e*.

`\GTSnegatedIncidence-MonochromeRelation`
`\GTSnegIncMonoRel`
`\GTSnegIncMonRel`
`\GTSnIMR`

This macro will draw a monochrome negated incidence symbol with relation spacing in math mode.
`$v\GTSnegatedIncidenceMonochromeRelation/e$`
or `$v\GTSnegIncMonoRel/e$`
or `$v\GTSnegIncMonRel/e$`
or `$v\GTSnIMR/e$`
will yield *v* ⫰̸ *e*.

`\GTSoutIncidenceMonochrome`
`\GTSoutIncMono`
`\GTSoutIncMon`
`\GTSoIM`

This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.
`A.\GTSoutIncidenceMonochrome/A.`
or `A.\GTSoutIncMono/A.`
or `A.\GTSoutIncMon/A.`
or `A.\GTSoIM/A.`
will yield A.⫰A.
`v \GTSoutIncidenceMonochrome/ e`
or `v \GTSoutIncMono/ e`
or `v \GTSoutIncMon/ e`

or `v \GTSoIM/ e`
will yield v e.
`$v \GTSoutIncidenceMonochrome/ e$`
or `$v \GTSoutIncMono/ e$`
or `$v \GTSoutIncMon/ e$`
or `$v \GTSoIM/ e$`
will yield v e.

`\GTSoutIncidenceMonochrome-Relation`
`\GTSoutIncMonoRel`
`\GTSoutIncMonRel`
`\GTSoIMR`

This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.
`$v\GTSoutIncidenceMonochromeRelation/e$`
or `$v\GTSoutIncMonoRel/e$`
or `$v\GTSoutIncMonRel/e$`
or `$v\GTSoIMR/e$`
will yield v e.

`\GTSnegatedOutIncidence-Monochrome`
`\GTSnegOutIncMono`
`\GTSnegOutIncMon`
`\GTSnOIM`

This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol.
`A.\GTSnegatedOutIncidenceMonochrome/A.`
or `A.\GTSnegOutIncMono/A.`
or `A.\GTSnegOutIncMon/A.`
or `A.\GTSnOIM/A.`
will yield A.A.
`v \GTSnegatedOutIncidenceMonochrome/ e`
or `v \GTSnegOutIncMono/ e`
or `v \GTSnegOutIncMon/ e`
or `v \GTSnOIM/ e`
will yield v e.
`$v \GTSnegatedOutIncidenceMonochrome/ e$`
or `$v \GTSnegOutIncMono/ e$`
or `$v \GTSnegOutIncMon/ e$`
or `$v \GTSnOIM/ e$`
will yield $v$$e$.

`\GTSnegatedOutIncidence-MonochromeRelation`
`\GTSnegOutIncMonoRel`
`\GTSnegOutIncMonRel`
`\GTSnOIMR`

This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.
`$v\GTSnegatedOutIncidenceMonochromeRelation/e$`
or `$v\GTSnegOutIncMonoRel/e$`
or `$v\GTSnegOutIncMonRel/e$`
or `$v\GTSnOIMR/e$`
will yield v e.

`\GTSinIncidenceMonochrome`
`\GTSinIncMono`
`\GTSinIncMon`
`\GTSiIM`

This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.
`A.\GTSinIncidenceMonochrome/A.`
or `A.\GTSinIncMono/A.`
or `A.\GTSinIncMon/A.`
or `A.\GTSiIM/A.`
will yield A.A.
`v \GTSinIncidenceMonochrome/ e`
or `v \GTSinIncMono/ e`
or `v \GTSinIncMon/ e`
or `v \GTSiIM/ e`
will yield v e.
`$v \GTSinIncidenceMonochrome/ e$`
or `$v \GTSinIncMono/ e$`
or `$v \GTSinIncMon/ e$`
or `$v \GTSiIM/ e$`
will yield $v$$e$.

`\GTSinIncidenceMonochrome-Relation`
`\GTSinIncMonoRel`
`\GTSinIncMonRel`
`\GTSiIMR`

This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.
`$v\GTSinIncidenceMonochromeRelation/e$`

or `$v\GTSinIncMonoRel/e$`
or `$v\GTSinIncMonRel/e$`
or `$v\GTSiIMR/e$`
will yield v e.

`\GTSnegatedInIncidenceMonochrome` `\GTSnegInIncMono` `\GTSnegInIncMon` `\GTSnIIM`

This macro will draw a monochrome negated directed incidence downward (in(ward) incidence) symbol.
`A.\GTSnegatedInIncidenceMonochrome/A.`
or `A.\GTSnegInIncMono/A.`
or `A.\GTSnegInIncMon/A.`
or `A.\GTSnIIM/A.`
will yield A.A.
`v \GTSnegatedInIncidenceMonochrome/ e`
or `v \GTSnegInIncMono/ e`
or `v \GTSnegInIncMon/ e`
or `v \GTSnIIM/ e`
will yield v e.
`$v \GTSnegatedInIncidenceMonochrome/ e$`
or `$v \GTSnegInIncMono/ e$`
or `$v \GTSnegInIncMon/ e$`
or `$v \GTSnIIM/ e$`
will yield $v$$e$.

`\GTSnegatedInIncidenceMonochromeRelation` `\GTSnegInIncMonoRel` `\GTSnegInIncMonRel` `\GTSnIIMR`

This macro will draw a monochrome negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.
`$v\GTSnegatedInIncidenceMonochromeRelation/e$`
or `$v\GTSnegInIncMonoRel/e$`
or `$v\GTSnegInIncMonRel/e$`
or `$v\GTSnIIMR/e$`
will yield v e.

`\GTSmetaNot` `\GTSmN`

This macro will draw a meta-not or relation-not symbol.
`A.\GTSmetaNot/A.`
or `A.\GTSmN/A.`
will yield A.A.
`u \GTSmetaNot/\GTSdAU/ v`
or `u \GTSmN/\GTSdAU/ v`
will yield u v.
`$u \GTSmetaNot/ \GTSdAU/ v$`
or `$u \GTSmN/ \GTSdAU/ v$`
will yield $u$$v$.

`\GTSmetaAnd` `\GTSmA`

This macro will draw a meta-and or relations-and symbol. This is your duty to check that all relations have same arity. It should be easy as long as you remain in the realm of graphs/2-structures/binary structures. We didn't use Tarski's notation, since square cup has taken the preemptive meaning of disjoint union. Similarly, doublewedge or other wedge symbols have taken various meanings, whilst we want an unambiguous symbol.
Look at the discussion after all logical symbols for less trivial uses and recommendations.
`A.\GTSmetaAnd/A.`
or `A.\GTSmA/A.`
will yield A.A.
`u \GTSdAD/\GTSmetaAnd/\GTSdAU/ v`
or `u \GTSdAD/\GTSmA/\GTSdAU/ v`
will yield u v.
`$u \GTSdAD/ \GTSmetaAnd/ \GTSdAU/ v$`
or `$u \GTSdAD/ \GTSmA/ \GTSdAU/ v$`
will yield $u$$v$.

`\GTSmetaAndRelation` `\GTSmetaAndRel` `\GTSmAR`

This macro will draw a meta-and symbol with relation spacing in math mode.
`$u \GTSdAD/\GTSmetaAndRelation/\GTSdAU/ v$`
or `$u \GTSdAD/\GTSmetaAndRel/\GTSdAU/ v$`
or `$u \GTSdAD/\GTSmAR/\GTSdAU/ v$`
will yield u v.

`\GTSmetaOr` `\GTSmO` This macro will draw a meta-or or relations-or symbol. This is your duty to check that all relations have same arity. It should be easy as long as you remain in the realm of graphs/2-structures/binary structures. We didn't use Tarski's notation, since square cup has taken the preemptive meaning of disjoint union. Similarly, doublewedge or other wedge symbols have taken various meanings, whilst we want an unambiguous symbol.
Look at the discussion after all logical symbols for less trivial uses and recommendations.
`A.\GTSmetaOr/A.`
or `A.\GTSmO/A.`
will yield A.$\overset{\mathrm{m}}{\vee}$A.
`u \GTSdAD/\GTSmetaOr/\GTSdAU/ v`
or `u \GTSdAD/\GTSmO/\GTSdAU/ v`
will yield u $\overset{\mathrm{m}}{\vee}$ v.
`$u \GTSdAD/ \GTSmetaOr/ \GTSdAU/ v$`
or `$u \GTSdAD/ \GTSmO/ \GTSdAU/ v$`
will yield $u\overset{\mathrm{m}}{\vee}v$.

`\GTSmetaOrRelation` `\GTSmetaOrRel` `\GTSmOR` This macro will draw a meta-or symbol with relation spacing in math mode.
`$u \GTSdAD/\GTSmetaOrRelation/\GTSdAU/ v$`
or `$u \GTSdAD/\GTSmetaOrRel/\GTSdAU/ v$`
or `$u \GTSdAD/\GTSmOR/\GTSdAU/ v$`
will yield $u \mathrel{\overset{\mathrm{m}}{\vee}} v$.

2.3 Good practices

As is customary in algebra and logic, the conjunction/multiplication can be replaced by concatenation, and has higher priority than disjunction. So the following formulas are equivalent; we recommend using the shortest one:
Tournament adjacency:
$\forall u, v, u \overset{\mathrm{m}}{\vee} v$
is prefered over
$\forall u, v, u \overset{\mathrm{m}}{\wedge}\overset{\mathrm{m}}{\vee}\overset{\mathrm{m}}{\wedge} v$,
or $\forall u, v, u\ (\overset{\mathrm{m}}{\wedge})\overset{\mathrm{m}}{\vee}(\overset{\mathrm{m}}{\wedge})\ v$.
Note the use of macros without Relation suffix above, and adding a single `\mathrel{}` around the composition of all relations:
`\mathrel{\GTSdAD/\GTSnDAU/\GTSmO/\GTSnDAD/\GTSdAU/}`.
This is the spacing practice that we recommend, although for very long compositions, it may be quite unreadable, and then spacing everything with Relation could/should be prefered.

We also recommend using brackets when the composition reflects all of the adjacencies between two vertices:
Tournament adjacency:
$\forall u, v, u\ [\overset{\mathrm{m}}{\vee}]\ v$.
Tournament adjacency with maybe some colored edge also, who knows?:
$\forall u, v, u \overset{\mathrm{m}}{\vee} v$.
Directed graph:
$\forall u, v, u\ [\overset{\mathrm{m}}{\vee}\overset{\mathrm{m}}{\vee}\overset{\mathrm{m}}{\vee}]\ v$.
Oriented graph:
$\forall u, v, u\ [\overset{\mathrm{m}}{\vee}\overset{\mathrm{m}}{\vee}]\ v$.

Finally, we recommend prefix notation without "big" symbol for a sequence of consecutive ors or ands (with parentheses if needed):
$\exists u, v, u\ [\overset{\mathrm{m}}{\vee}]\ v$.
$\exists v, e, v\ [\overset{\mathrm{m}}{\vee}]\ e$.

3 Implementation

```
⟨*package⟩
\RequirePackage{amsmath}
\RequirePackage{amssymb}
\RequirePackage{graphicx}
\RequirePackage{keyval}
\RequirePackage{fdsymbol}
\RequirePackage{lua-unicode-math}
```

```
\RequirePackage{xcolor}

\def\GTS@mainColor{black}
\def\GTS@secondColor{gray}
\def\GTS@mainColorBackup{black}
\def\GTS@secondColorBackup{gray}
\def\GTS@drawSlash{}
\def\GTS@drawUpArrow{}
\def\GTS@drawDownArrow{}
```

`\GTS@newcommand` This macro will enforce that the other macros are called with a trailing slash, avoiding problems with spaces after macro call.
Arguments:
{⟨*commandname*⟩} The name of the command to create.
{⟨*commandcode*⟩} The code of the command to create.

```
\@ifdefinable{\GTS@newcommand}{\def\GTS@newcommand#1#2{%
  \@ifdefinable{#1}{\def#1/{#2}}
}}
```

`\GTS@newprotectedcommand` This macro will enforce that the other macros are called with a trailing slash, avoiding problems with spaces after macro call.
Arguments:
{⟨*commandname*⟩} The name of the command to create.
{⟨*commandcode*⟩} The code of the command to create.

```
\@ifdefinable{\GTS@newprotectedcommand}{\def\GTS@newprotectedcommand#1#2{%
  \@ifdefinable{#1}{\protected\def#1/{#2}}
}}
```

`\GTSsetMainColor` This macro will set the main color used in GraphTheorySymbol. This color has initial value "black".
Argument:
{⟨*color*⟩} The color that must be used by GraphTheorySymbol afterward.

```
\newcommand{\GTSsetMainColor}[1]{%
\edef\GTS@mainColorTemp{#1}%
\edef\GTS@mainColorBackup{\GTS@mainColor}%
\edef\GTS@mainColor{\GTS@mainColorTemp}%
}
```

`\GTSsetSecondColor` This macro will set the second color used in GraphTheorySymbol. This color has initial value "gray".
Argument:
{⟨*color*⟩} The color that must be used by GraphTheorySymbol afterward.

```
\newcommand{\GTSsetSecondColor}[1]{%
\edef\GTS@secondColorTemp{#1}%
\edef\GTS@secondColorBackup{\GTS@secondColor}%
\edef\GTS@secondColor{\GTS@secondColorTemp}%
}
```

`\GTS@a` This macro will draw any adjacency symbol given the correct parameters. Its use is purely technical. See the macros derived from it and `\GTS@b`: `\GTSxadjacency`, `\GTSxadj`, and `\GTSxa`.
Arguments:
{⟨*mainColor*⟩} If non-empty, this color will be used as main color.
{⟨*secondColor*⟩} If non-empty, this color will be used as second color.
{⟨*drawSlash*⟩} Non-empty implies that a negation will be drawn.
{⟨*drawUpArrow*⟩} Non-empty implies that an up-arrow will be drawn.
{⟨*drawDownArrow*⟩} Non-empty implies that a down-arrow will be drawn.

```
\newcommand{\GTS@a}[5]{%
\if#1\empty\else%
\GTSsetMainColor{#1}%
\fi%
\if#2\empty\else%
\GTSsetSecondColor{#2}%
```

```
\fi%
\text{\ensuremath{%
\raisebox{-0.32ex}{\scalebox{1.5}[1.5]{%
%
\raisebox{0.7ex}{\scalebox{0.7}[0.7]{%
$\mathcolor{\GTS@secondColor}{\circ}$%
}}%
\hspace{-0.39ex}%
\raisebox{1.05ex}{\scalebox{0.1}[0.1]{%
$\mathcolor{\GTS@secondColor}{\bullet}$%
}}%
\hspace{-0.12ex}%
\raisebox{0.11ex}{\scalebox{0.1}[0.73]{%
$\mathcolor{\GTS@mainColor}{\smallblacksquare}$%
}}%
\hspace{-0.40ex}%
\raisebox{-0.3ex}{\scalebox{0.7}[0.7]{%
$\mathcolor{\GTS@secondColor}{\circ}$%
}}%
\hspace{-0.39ex}%
\raisebox{0.05ex}{\scalebox{0.1}[0.1]{%
$\mathcolor{\GTS@secondColor}{\bullet}$%
}}%
\hspace{0.27ex}%
```

if GTS@drawUpArrow empty else

```
\if#4\empty\else%
\hspace{-0.65ex}%
\raisebox{-0.18ex}{\scalebox{0.5}[0.5]{%
\textcolor{\GTS@mainColor}{\textasciicircum}%
}}%
\fi%
```

if GTS@drawDownArrow empty else

```
\if#5\empty\else%
\hspace{-0.65ex}%
\raisebox{1.37ex}{\scalebox{0.5}[-0.5]{%
\textcolor{\GTS@mainColor}{\textasciicircum}%
}}%
\fi%
```

if GTS@drawSlash empty else

```
\if#3\empty\else%
\hspace{-0.65ex}%
\raisebox{0.25ex}{\scalebox{0.6}[0.6]{%
\textcolor{black}{/}%
}}%
\fi%
%
}}%
}}%
\if#1\empty\else%
\GTSsetMainColor{\GTS@mainColorBackup}%
\fi%
\if#2\empty\else%
\GTSsetSecondColor{\GTS@secondColorBackup}%
\fi%
}
```

`\GTS@b` This macro will handle wrapping in mathematical relation spacing if needed. Its use is purely technical. See the macros derived from it and `\GTS@a`: `\GTSxadjacency`, `\GTSxadj`, and `\GTSxa`.

Arguments:

{⟨*mainColor*⟩} If non-empty, this color will be used as main color.

{⟨*secondColor*⟩} If non-empty, this color will be used as second color.

{⟨*drawSlash*⟩} Non-empty implies that a negation will be drawn.

{⟨*drawUpArrow*⟩} Non-empty implies that an up-arrow will be drawn.

{⟨*drawDownArrow*⟩} Non-empty implies that a down-arrow will be drawn.
{⟨*relation*⟩} Non-empty implies that relation spacing will be added.

```
\newcommand{\GTS@b}[6]{%
\if#61%
\mathrel{\GTS@a{#1}{#2}{#3}{#4}{#5}}%
\else%
\GTS@a{#1}{#2}{#3}{#4}{#5}%
\fi%
}
```

`\GTSxa` `\GTSxadj` `\GTSxadjacency`

Versatile macros for adjacency that can be given options.
Argument:
[⟨*optionsList*⟩] Sets the following options if not empty:
mainColor=color1 or mainC=color1 or mColor=color1 or mC=color1,
secondColor=color2 or secondC=color2 or sColor=color2 or sC=color2,
monochrome or mono or mon or m (equivalent to mC=black,sC=black),
negated or not or n,
up or u,
down or d,
relation or rel or r.

```
\define@key{GTSxa}{mainColor}{\def\GTS@@mainColor{#1}}
\define@key{GTSxa}{mainC}{\def\GTS@@mainColor{#1}}
\define@key{GTSxa}{mColor}{\def\GTS@@mainColor{#1}}
\define@key{GTSxa}{mC}{\def\GTS@@mainColor{#1}}
\define@key{GTSxa}{secondColor}{\def\GTS@@secondColor{#1}}
\define@key{GTSxa}{secondC}{\def\GTS@@secondColor{#1}}
\define@key{GTSxa}{sColor}{\def\GTS@@secondColor{#1}}
\define@key{GTSxa}{sC}{\def\GTS@@secondColor{#1}}
\define@key{GTSxa}{monochrome}[1]%
{\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
\define@key{GTSxa}{mono}[1]%
{\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
\define@key{GTSxa}{mon}[1]%
{\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
\define@key{GTSxa}{m}[1]%
{\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
\define@key{GTSxa}{negated}[1]{\def\GTS@@drawSlash{1}}
\define@key{GTSxa}{not}[1]{\def\GTS@@drawSlash{1}}
\define@key{GTSxa}{n}[1]{\def\GTS@@drawSlash{1}}
\define@key{GTSxa}{up}[1]{\def\GTS@@drawUpArrow{1}}
\define@key{GTSxa}{u}[1]{\def\GTS@@drawUpArrow{1}}
\define@key{GTSxa}{down}[1]{\def\GTS@@drawDownArrow{1}}
\define@key{GTSxa}{d}[1]{\def\GTS@@drawDownArrow{1}}
\define@key{GTSxa}{relation}[1]{\def\GTS@@relation{1}}
\define@key{GTSxa}{rel}[1]{\def\GTS@@relation{1}}
\define@key{GTSxa}{r}[1]{\def\GTS@@relation{1}}

\newcommand{\GTSxa}[1][]{%
\def\GTS@@mainColor{}%
\def\GTS@@secondColor{}%
\def\GTS@@drawSlash{}%
\def\GTS@@drawUpArrow{}%
\def\GTS@@drawDownArrow{}%
\def\GTS@@relation{}%
\setkeys{GTSxa}{#1}%
\GTS@b{\GTS@@mainColor}{\GTS@@secondColor}{\GTS@@drawSlash}%
{\GTS@@drawUpArrow}{\GTS@@drawDownArrow}{\GTS@@relation}%
}

\newcommand{\GTSxadj}[1][]{\GTSxa[#1]}

\newcommand{\GTSxadjacency}[1][]{\GTSxa[#1]}
```

`\GTSadjacency` `\GTSadj` `\GTSa`

This macro will draw an adjacency symbol.

```
\GTS@newcommand{\GTSadjacency}{\GTS@a{}{}{}{}{}}
```

```
\GTS@newcommand{\GTSadj}{\GTS@a{}{}{}{}{}}
\GTS@newcommand{\GTSa}{\GTS@a{}{}{}{}{}}
```

`\GTSadjacencyRelation` `\GTSadjRel` `\GTSaR`

This macro will draw an adjacency symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSadjacencyRelation}{\mathrel{\GTSa/}}
\GTS@newcommand{\GTSadjRel}{\mathrel{\GTSa/}}
\GTS@newcommand{\GTSaR}{\mathrel{\GTSa/}}
```

`\GTSunadjacency` `\GTSunadj` `\GTSu`

This macro will draw an unadjacency symbol.

```
\GTS@newcommand{\GTSunadjacency}{\GTS@a{}{}{1}{}{}}
\GTS@newcommand{\GTSunadj}{\GTS@a{}{}{1}{}{}}
\GTS@newcommand{\GTSu}{\GTS@a{}{}{1}{}{}}
```

`\GTSunadjacencyRelation` `\GTSunadjRel` `\GTSuR`

This macro will draw an unadjacency symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSunadjacencyRelation}{\mathrel{\GTSu/}}
\GTS@newcommand{\GTSunadjRel}{\mathrel{\GTSu/}}
\GTS@newcommand{\GTSuR}{\mathrel{\GTSu/}}
```

`\GTSdirectedAdjacencyUp` `\GTSdirAdjUp` `\GTSdAU`

This macro will draw a directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

```
\GTS@newcommand{\GTSdirectedAdjacencyUp}{\GTS@a{}{}{}{1}{}}
\GTS@newcommand{\GTSdirAdjUp}{\GTS@a{}{}{}{1}{}}
\GTS@newcommand{\GTSdAU}{\GTS@a{}{}{}{1}{}}
```

`\GTSdirectedAdjacencyUp-Relation` `\GTSdirAdjUpRel` `\GTSdAUR`

This macro will draw a directed adjacency upward symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSdirectedAdjacencyUpRelation}{\mathrel{\GTSdAU/}}
\GTS@newcommand{\GTSdirAdjUpRel}{\mathrel{\GTSdAU/}}
\GTS@newcommand{\GTSdAUR}{\mathrel{\GTSdAU/}}
```

`\GTSnegatedDirected-AdjacencyUp` `\GTSnegDirAdjUp` `\GTSnDAU`

This macro will draw a negated directed adjacency upward symbol.

```
\GTS@newcommand{\GTSnegatedDirectedAdjacencyUp}{\GTS@a{}{}{1}{1}{}}
\GTS@newcommand{\GTSnegDirAdjUp}{\GTS@a{}{}{1}{1}{}}
\GTS@newcommand{\GTSnDAU}{\GTS@a{}{}{1}{1}{}}
```

`\GTSnegatedDirected-AdjacencyUpRelation` `\GTSnegDirAdjUpRel` `\GTSnDAUR`

This macro will draw a negated directed adjacency upward symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSnegatedDirectedAdjacencyUpRelation}{%
\mathrel{\GTSnDAU/}%
}
\GTS@newcommand{\GTSnegDirAdjUpRel}{\mathrel{\GTSnDAU/}}
\GTS@newcommand{\GTSnDAUR}{\mathrel{\GTSnDAU/}}
```

`\GTSdirectedAdjacencyDown` `\GTSdirAdjDown` `\GTSdAD`

This macro will draw a directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.

```
\GTS@newcommand{\GTSdirectedAdjacencyDown}{\GTS@a{}{}{}{}{1}}
\GTS@newcommand{\GTSdirAdjDown}{\GTS@a{}{}{}{}{1}}
\GTS@newcommand{\GTSdAD}{\GTS@a{}{}{}{}{1}}
```

`\GTSdirectedAdjacencyDown-Relation` `\GTSdirAdjDownRel` `\GTSdADR`

This macro will draw a directed adjacency downward symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSdirectedAdjacencyDownRelation}{\mathrel{\GTSdAD/}}
\GTS@newcommand{\GTSdirAdjDownRel}{\mathrel{\GTSdAD/}}
\GTS@newcommand{\GTSdADR}{\mathrel{\GTSdAD/}}
```

`\GTSnegatedDirected-AdjacencyDown` `\GTSnegDirAdjDown` `\GTSnDAD`

This macro will draw a negated directed adjacency downward symbol.

```
\GTS@newcommand{\GTSnegatedDirectedAdjacencyDown}{\GTS@a{}{}{1}{}{1}}
\GTS@newcommand{\GTSnegDirAdjDown}{\GTS@a{}{}{1}{}{1}}
\GTS@newcommand{\GTSnDAD}{\GTS@a{}{}{1}{}{1}}
```

`\GTSnegatedDirectedAdjacencyDownRelation` `\GTSnegDirAdjDownRel` `\GTSnDADR`

This macro will draw a negated directed adjacency downward symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSnegatedDirectedAdjacencyDownRelation}{%
\mathrel{\GTSnDAD/}%
}
\GTS@newcommand{\GTSnegDirAdjDownRel}{\mathrel{\GTSnDAD/}}
\GTS@newcommand{\GTSnDADR}{\mathrel{\GTSnDAD/}}
```

`\GTS@i`

This macro will draw any incidence symbol given the correct parameters. Its use is purely technical. See the macros derived from it and `\GTS@j`: `\GTSxincidence`, `\GTSxinc`, and `\GTSxi`. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.
Arguments:
{⟨*mainColor*⟩} If non-empty, this color will be used as main color.
{⟨*secondColor*⟩} If non-empty, this color will be used as second color.
{⟨*drawSlash*⟩} Non-empty implies that a negation will be drawn.
{⟨*drawUpArrow*⟩} Non-empty implies that an up-arrow will be drawn.
{⟨*drawDownArrow*⟩} Non-empty implies that a down-arrow will be drawn.

```
\newcommand{\GTS@i}[5]{%
\if#1\empty\else%
\GTSsetMainColor{#1}%
\fi%
\if#2\empty\else%
\GTSsetSecondColor{#2}%
\fi%
\text{\ensuremath{%
\raisebox{-0.22ex}{\scalebox{2.05}[2.05]{%
%
\raisebox{-0.3ex}{\scalebox{0.7}[0.7]{%
$\mathcolor{\GTS@secondColor}{\circ}$%
}}%
\hspace{-0.39ex}%
\raisebox{0.05ex}{\scalebox{0.1}[0.1]{%
$\mathcolor{\GTS@secondColor}{\bullet}$%
}}%
\hspace{-0.12ex}%
\raisebox{0.11ex}{\scalebox{0.1}[0.73]{%
$\mathcolor{\GTS@secondColor}{\smallblacksquare}$%
}}%
\hspace{-0.18ex}%
\raisebox{0.15ex}{\scalebox{0.2}[0.2]{%
$\mathcolor{\GTS@mainColor}{\smallblacksquare}$%
}}%
\hspace{0.2ex}%
```

if GTS@drawUpArrow empty else

```
\if#4\empty\else%
\hspace{-0.65ex}%
\raisebox{-0.18ex}{\scalebox{0.5}[0.5]{%
\textcolor{\GTS@secondColor}{\textasciicircum}%
}}%
\fi%
```

if GTS@drawDownArrow empty else

```
\if#5\empty\else%
\hspace{-0.65ex}%
\raisebox{1.37ex}{\scalebox{0.5}[-0.5]{%
\textcolor{\GTS@secondColor}{\textasciicircum}%
}}%
\fi%
```

if GTS@drawSlash empty else

```
\if#3\empty\else%
\hspace{-0.65ex}%
\raisebox{0.1ex}{\scalebox{0.5}[0.5]{%
\textcolor{black}{/}%
```

```
}}%
\fi%
%
}}%
}}%
\if#1\empty\else%
\GTSsetMainColor{\GTS@mainColorBackup}%
\fi%
\if#2\empty\else%
\GTSsetSecondColor{\GTS@secondColorBackup}%
\fi%
}
```

`\GTS@j` This macro will handle wrapping in mathematical relation spacing if needed. Its use is purely technical. See the macros derived from it and `\GTS@i`: `\GTSxincidence`, `\GTSxinc`, and `\GTSxi`.
Arguments:
{⟨*mainColor*⟩} If non-empty, this color will be used as main color.
{⟨*secondColor*⟩} If non-empty, this color will be used as second color.
{⟨*drawSlash*⟩} Non-empty implies that a negation will be drawn.
{⟨*drawUpArrow*⟩} Non-empty implies that an up-arrow will be drawn.
{⟨*drawDownArrow*⟩} Non-empty implies that a down-arrow will be drawn.
{⟨*relation*⟩} Non-empty implies that relation spacing will be added.

```
\newcommand{\GTS@j}[6]{%
\if#61%
\mathrel{\GTS@i{#1}{#2}{#3}{#4}{#5}}%
\else%
\GTS@i{#1}{#2}{#3}{#4}{#5}%
\fi%
}
```

`\GTSxi` `\GTSxinc` `\GTSxincidence` Versatile macros for incidence that can be given options.
Argument:
[⟨*optionsList*⟩] Sets the following options if not empty:
mainColor=color1 or mainC=color1 or mColor=color1 or mC=color1,
secondColor=color2 or secondC=color2 or sColor=color2 or sC=color2,
monochrome or mono or mon or m (equivalent to mC=black,sC=black),
negated or not or n,
in or i,
out or o,
relation or rel or r.

```
\define@key{GTSxi}{mainColor}{\def\GTS@@mainColor{#1}}
\define@key{GTSxi}{mainC}{\def\GTS@@mainColor{#1}}
\define@key{GTSxi}{mColor}{\def\GTS@@mainColor{#1}}
\define@key{GTSxi}{mC}{\def\GTS@@mainColor{#1}}
\define@key{GTSxi}{secondColor}{\def\GTS@@secondColor{#1}}
\define@key{GTSxi}{secondC}{\def\GTS@@secondColor{#1}}
\define@key{GTSxi}{sColor}{\def\GTS@@secondColor{#1}}
\define@key{GTSxi}{sC}{\def\GTS@@secondColor{#1}}
\define@key{GTSxi}{monochrome}[1]%
{\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
\define@key{GTSxi}{mono}[1]%
{\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
\define@key{GTSxi}{mon}[1]%
{\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
\define@key{GTSxi}{m}[1]%
{\def\GTS@@mainColor{black}\def\GTS@@secondColor{black}}
\define@key{GTSxi}{negated}[1]{\def\GTS@@drawSlash{1}}
\define@key{GTSxi}{not}[1]{\def\GTS@@drawSlash{1}}
\define@key{GTSxi}{n}[1]{\def\GTS@@drawSlash{1}}
\define@key{GTSxi}{out}[1]{\def\GTS@@drawUpArrow{1}}
\define@key{GTSxi}{o}[1]{\def\GTS@@drawUpArrow{1}}
\define@key{GTSxi}{in}[1]{\def\GTS@@drawDownArrow{1}}
\define@key{GTSxi}{i}[1]{\def\GTS@@drawDownArrow{1}}
```

```
\define@key{GTSxi}{relation}[1]{\def\GTS@@relation{1}}
\define@key{GTSxi}{rel}[1]{\def\GTS@@relation{1}}
\define@key{GTSxi}{r}[1]{\def\GTS@@relation{1}}

\newcommand{\GTSxi}[1][]{%
\def\GTS@@mainColor{}%
\def\GTS@@secondColor{}%
\def\GTS@@drawSlash{}%
\def\GTS@@drawUpArrow{}%
\def\GTS@@drawDownArrow{}%
\def\GTS@@relation{}%
\setkeys{GTSxi}{#1}%
\GTS@j{\GTS@@mainColor}{\GTS@@secondColor}{\GTS@@drawSlash}%
{\GTS@@drawUpArrow}{\GTS@@drawDownArrow}{\GTS@@relation}%
}

\newcommand{\GTSxinc}[1][]{\GTSxi[#1]}

\newcommand{\GTSxincidence}[1][]{\GTSxi[#1]}
```

`\GTSincidence` `\GTSinc` `\GTSi`

This macro will draw an incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

```
\GTS@newcommand{\GTSincidence}{\GTS@i{}{}{}{}{}}
\GTS@newcommand{\GTSinc}{\GTS@i{}{}{}{}{}}
\GTS@newcommand{\GTSi}{\GTS@i{}{}{}{}{}}
```

`\GTSincidenceRelation` `\GTSincRel` `\GTSiR`

This macro will draw an incidence symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSincidenceRelation}{\mathrel{\GTSi/}}
\GTS@newcommand{\GTSincRel}{\mathrel{\GTSi/}}
\GTS@newcommand{\GTSiR}{\mathrel{\GTSi/}}
```

`\GTSnegatedIncidence` `\GTSnegInc` `\GTSnI`

This macro will draw a negated incidence symbol.

```
\GTS@newcommand{\GTSnegatedIncidence}{\GTS@i{}{}{1}{}{}}
\GTS@newcommand{\GTSnegInc}{\GTS@i{}{}{1}{}{}}
\GTS@newcommand{\GTSnI}{\GTS@i{}{}{1}{}{}}
```

`\GTSnegatedIncidence-Relation` `\GTSnegIncRel` `\GTSnIR`

This macro will draw a negated incidence symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSnegatedIncidenceRelation}{\mathrel{\GTSnI/}}
\GTS@newcommand{\GTSnegIncRel}{\mathrel{\GTSnI/}}
\GTS@newcommand{\GTSnIR}{\mathrel{\GTSnI/}}
```

`\GTSoutIncidence` `\GTSoutInc` `\GTSoI`

This macro will draw a directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

```
\GTS@newcommand{\GTSoutIncidence}{\GTS@i{}{}{}{1}{}}
\GTS@newcommand{\GTSoutInc}{\GTS@i{}{}{}{1}{}}
\GTS@newcommand{\GTSoI}{\GTS@i{}{}{}{1}{}}
```

`\GTSoutIncidenceRelation` `\GTSoutIncRel` `\GTSoIR`

This macro will draw a directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSoutIncidenceRelation}{\mathrel{\GTSoI/}}
\GTS@newcommand{\GTSoutIncRel}{\mathrel{\GTSoI/}}
\GTS@newcommand{\GTSoIR}{\mathrel{\GTSoI/}}
```

`\GTSnegatedOutIncidence` `\GTSnegOutInc` `\GTSnOI`

This macro will draw a negated directed incidence upward (out(ward) incidence) symbol.

```
\GTS@newcommand{\GTSnegatedOutIncidence}{\GTS@i{}{}{1}{1}{}}
\GTS@newcommand{\GTSnegOutInc}{\GTS@i{}{}{1}{1}{}}
\GTS@newcommand{\GTSnOI}{\GTS@i{}{}{1}{1}{}}
```

`\GTSnegatedOutIncidence-Relation` `\GTSnegOutIncRel` `\GTSnOIR`

This macro will draw a negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSnegatedOutIncidenceRelation}{\mathrel{\GTSnOI/}}
\GTS@newcommand{\GTSnegOutIncRel}{\mathrel{\GTSnOI/}}
\GTS@newcommand{\GTSnOIR}{\mathrel{\GTSnOI/}}
```

`\GTSinIncidence` `\GTSinInc` `\GTSiI`

This macro will draw a directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

```
\GTS@newcommand{\GTSinIncidence}{\GTS@i{}{}{}{}{1}}
\GTS@newcommand{\GTSinInc}{\GTS@i{}{}{}{}{1}}
\GTS@newcommand{\GTSiI}{\GTS@i{}{}{}{}{1}}
```

`\GTSinIncidenceRelation` `\GTSinIncRel` `\GTSiIR`

This macro will draw a directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSinIncidenceRelation}{\mathrel{\GTSiI/}}
\GTS@newcommand{\GTSinIncRel}{\mathrel{\GTSiI/}}
\GTS@newcommand{\GTSiIR}{\mathrel{\GTSiI/}}
```

`\GTSnegatedInIncidence` `\GTSnegInInc` `\GTSnII`

This macro will draw a negated directed incidence downward (in(ward) incidence) symbol.

```
\GTS@newcommand{\GTSnegatedInIncidence}{\GTS@i{}{}{1}{}{1}}
\GTS@newcommand{\GTSnegInInc}{\GTS@i{}{}{1}{}{1}}
\GTS@newcommand{\GTSnII}{\GTS@i{}{}{1}{}{1}}
```

`\GTSnegatedInIncidence-Relation` `\GTSnegInIncRel` `\GTSnIIR`

This macro will draw a negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSnegatedInIncidenceRelation}{\mathrel{\GTSnII/}}
\GTS@newcommand{\GTSnegInIncRel}{\mathrel{\GTSnII/}}
\GTS@newcommand{\GTSnIIR}{\mathrel{\GTSnII/}}
```

`\GTSadjacencyMonochrome` `\GTSadjMono` `\GTSadjMon` `\GTSaM`

This macro will draw a monochrome adjacency symbol.

```
\GTS@newcommand{\GTSadjacencyMonochrome}{\GTS@a{black}{black}{}{}{}}
\GTS@newcommand{\GTSadjMono}{\GTS@a{black}{black}{}{}{}}
\GTS@newcommand{\GTSadjMon}{\GTS@a{black}{black}{}{}{}}
\GTS@newcommand{\GTSaM}{\GTS@a{black}{black}{}{}{}}
```

`\GTSadjacencyMonochrome-Relation` `\GTSadjMonoRel` `\GTSadjMonRel` `\GTSaMR`

This macro will draw a monochrome adjacency symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSadjacencyMonochromeRelation}{\mathrel{\GTSaM/}}
\GTS@newcommand{\GTSadjMonoRel}{\mathrel{\GTSaM/}}
\GTS@newcommand{\GTSadjMonRel}{\mathrel{\GTSaM/}}
\GTS@newcommand{\GTSaMR}{\mathrel{\GTSaM/}}
```

`\GTSunadjacencyMonochrome` `\GTSunadjMono` `\GTSunadjMon` `\GTSuM`

This macro will draw a monochrome unadjacency symbol.

```
\GTS@newcommand{\GTSunadjacencyMonochrome}{\GTS@a{black}{black}{1}{}{}}
\GTS@newcommand{\GTSunadjMono}{\GTS@a{black}{black}{1}{}{}}
\GTS@newcommand{\GTSunadjMon}{\GTS@a{black}{black}{1}{}{}}
\GTS@newcommand{\GTSuM}{\GTS@a{black}{black}{1}{}{}}
```

`\GTSunadjacencyMonochrome-Relation` `\GTSunadjMonoRel` `\GTSunadjMonRel` `\GTSuMR`

This macro will draw a monochrome unadjacency symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSunadjacencyMonochromeRelation}{\mathrel{\GTSuM/}}
\GTS@newcommand{\GTSunadjMonoRel}{\mathrel{\GTSuM/}}
\GTS@newcommand{\GTSunadjMonRel}{\mathrel{\GTSuM/}}
\GTS@newcommand{\GTSuMR}{\mathrel{\GTSuM/}}
```

`\GTSdirectedAdjacencyUp-Monochrome` `\GTSdirAdjUpMono` `\GTSdirAdjUpMon` `\GTSdAUM`

This macro will draw a monochrome directed adjacency upward symbol. The operand on the left of the symbol should be considered the source of the arc; the operand on the right of the symbol should be considered the target of the arc.

```
\GTS@newcommand{\GTSdirectedAdjacencyUpMonochrome}{%
\GTS@a{black}{black}{}{1}{}%
}
\GTS@newcommand{\GTSdirAdjUpMono}{\GTS@a{black}{black}{}{1}{}}
\GTS@newcommand{\GTSdirAdjUpMon}{\GTS@a{black}{black}{}{1}{}}
\GTS@newcommand{\GTSdAUM}{\GTS@a{black}{black}{}{1}{}}
```

`\GTSdirectedAdjacencyUp-MonochromeRelation` `\GTSdirAdjUpMonoRel` `\GTSdirAdjUpMonRel` `\GTSdAUMR`

This macro will draw a monochrome directed adjacency upward symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSdirectedAdjacencyUpMonochromeRelation}{%
\mathrel{\GTSdAUM/}%
}
```

```
\GTS@newcommand{\GTSdirAdjUpMonoRel}{\mathrel{\GTSdAUM/}}
\GTS@newcommand{\GTSdirAdjUpMonRel}{\mathrel{\GTSdAUM/}}
\GTS@newcommand{\GTSdAUMR}{\mathrel{\GTSdAUM/}}
```

\GTSnegatedDirected-AdjacencyUpMonochrome \GTSnegDirAdjUpMono \GTSnegDirAdjUpMon \GTSnDAUM

This macro will draw a monochrome negated directed adjacency upward symbol.

```
\GTS@newcommand{\GTSnegatedDirectedAdjacencyUpMonochrome}{%
\GTS@a{black}{black}{1}{1}{}%
}
\GTS@newcommand{\GTSnegDirAdjUpMono}{\GTS@a{black}{black}{1}{1}{}}
\GTS@newcommand{\GTSnegDirAdjUpMon}{\GTS@a{black}{black}{1}{1}{}}
\GTS@newcommand{\GTSnDAUM}{\GTS@a{black}{black}{1}{1}{}}
```

\GTSnegatedDirected-AdjacencyUpMonochrome-Relation \GTSnegDirAdjUpMonRel \GTSnegDirAdjUpMonoRel \GTSnDAUMR

This macro will draw a monochrome negated directed adjacency upward symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSnegatedDirectedAdjacencyUpMonochromeRelation}{%
\mathrel{\GTSnDAUM/}%
}
\GTS@newcommand{\GTSnegDirAdjUpMonoRel}{\mathrel{\GTSnDAUM/}}
\GTS@newcommand{\GTSnegDirAdjUpMonRel}{\mathrel{\GTSnDAUM/}}
\GTS@newcommand{\GTSnDAUMR}{\mathrel{\GTSnDAUM/}}
```

\GTSdirectedAdjacencyDown-Monochrome \GTSdirAdjDownMono \GTSdirAdjDownMon \GTSdADM

This macro will draw a monochrome directed adjacency downward symbol. The operand on the left of the symbol should be considered the target of the arc; the operand on the right of the symbol should be considered the source of the arc.

```
\GTS@newcommand{\GTSdirectedAdjacencyDownMonochrome}{%
\GTS@a{black}{black}{}{}{1}%
}
\GTS@newcommand{\GTSdirAdjDownMono}{\GTS@a{black}{black}{}{}{1}}
\GTS@newcommand{\GTSdirAdjDownMon}{\GTS@a{black}{black}{}{}{1}}
\GTS@newcommand{\GTSdADM}{\GTS@a{black}{black}{}{}{1}}
```

\GTSdirectedAdjacencyDown-MonochromeRelation \GTSdirAdjDownMonoRel \GTSdirAdjDownMonRel \GTSdADMR

This macro will draw a monochrome directed adjacency downward symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSdirectedAdjacencyDownMonochromeRelation}{%
\mathrel{\GTSdADM/}%
}
\GTS@newcommand{\GTSdirAdjDownMonoRel}{\mathrel{\GTSdADM/}}
\GTS@newcommand{\GTSdirAdjDownMonRel}{\mathrel{\GTSdADM/}}
\GTS@newcommand{\GTSdADMR}{\mathrel{\GTSdADM/}}
```

\GTSnegatedDirected-AdjacencyDownMonochrome \GTSnegDirAdjDownMono \GTSnegDirAdjDownMon \GTSnDADM

This macro will draw a monochrome negated directed adjacency downward symbol.

```
\GTS@newcommand{\GTSnegatedDirectedAdjacencyDownMonochrome}{%
\GTS@a{black}{black}{1}{}{1}%
}
\GTS@newcommand{\GTSnegDirAdjDownMono}{\GTS@a{black}{black}{1}{}{1}}
\GTS@newcommand{\GTSnegDirAdjDownMon}{\GTS@a{black}{black}{1}{}{1}}
\GTS@newcommand{\GTSnDADM}{\GTS@a{black}{black}{1}{}{1}}
```

\GTSnegatedDirected-AdjacencyDownMonochrome-Relation \GTSnegDirAdjDownMonoRel \GTSnegDirAdjDownMonRel \GTSnDADMR

This macro will draw a monochrome negated directed adjacency downward symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSnegatedDirectedAdjacencyDownMonochromeRelation}{%
\mathrel{\GTSnDADM/}%
}
\GTS@newcommand{\GTSnegDirAdjDownMonoRel}{\mathrel{\GTSnDADM/}}
\GTS@newcommand{\GTSnegDirAdjDownMonRel}{\mathrel{\GTSnDADM/}}
\GTS@newcommand{\GTSnDADMR}{\mathrel{\GTSnDADM/}}
```

\GTSincidenceMonochrome \GTSincMono \GTSincMon \GTSiM

This macro will draw a monochrome incidence symbol. The operand on the left of the symbol should be considered the vertex to which the edge/the operand on the right of the symbol is incident.

```
\GTS@newcommand{\GTSincidenceMonochrome}{\GTS@i{black}{black}{}{}{}}
\GTS@newcommand{\GTSincMono}{\GTS@i{black}{black}{}{}{}}
\GTS@newcommand{\GTSincMon}{\GTS@i{black}{black}{}{}{}}
\GTS@newcommand{\GTSiM}{\GTS@i{black}{black}{}{}{}}
```

`\GTSincidenceMonochromeRelation` `\GTSincMonoRel` `\GTSincMonRel` `\GTSiMR`

This macro will draw a monochrome incidence symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSincidenceMonochromeRelation}{\mathrel{\GTSiM/}}
\GTS@newcommand{\GTSincMonoRel}{\mathrel{\GTSiM/}}
\GTS@newcommand{\GTSincMonRel}{\mathrel{\GTSiM/}}
\GTS@newcommand{\GTSiMR}{\mathrel{\GTSiM/}}
```

`\GTSnegatedIncidenceMonochrome` `\GTSnegIncMono` `\GTSnegIncMon` `\GTSnIM`

This macro will draw a monochrome negated incidence symbol.

```
\GTS@newcommand{\GTSnegatedIncidenceMonochrome}{%
\GTS@i{black}{black}{1}{}{}%
}
\GTS@newcommand{\GTSnegIncMono}{\GTS@i{black}{black}{1}{}{}}
\GTS@newcommand{\GTSnegIncMon}{\GTS@i{black}{black}{1}{}{}}
\GTS@newcommand{\GTSnIM}{\GTS@i{black}{black}{1}{}{}}
```

`\GTSnegatedIncidenceMonochromeRelation` `\GTSnegIncMonoRel` `\GTSnegIncMonRel` `\GTSnIMR`

This macro will draw a monochrome negated incidence symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSnegatedIncidenceMonochromeRelation}{\mathrel{\GTSnIM/}}
\GTS@newcommand{\GTSnegIncMonoRel}{\mathrel{\GTSnIM/}}
\GTS@newcommand{\GTSnegIncMonRel}{\mathrel{\GTSnIM/}}
\GTS@newcommand{\GTSnIMR}{\mathrel{\GTSnIM/}}
```

`\GTSoutIncidenceMonochrome` `\GTSoutIncMono` `\GTSoutIncMon` `\GTSoIM`

This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

```
\GTS@newcommand{\GTSoutIncidenceMonochrome}{\GTS@i{black}{black}{}{1}{}}
\GTS@newcommand{\GTSoutIncMono}{\GTS@i{black}{black}{}{1}{}}
\GTS@newcommand{\GTSoutIncMon}{\GTS@i{black}{black}{}{1}{}}
\GTS@newcommand{\GTSoIM}{\GTS@i{black}{black}{}{1}{}}
```

`\GTSoutIncidenceMonochromeRelation` `\GTSoutIncMonoRel` `\GTSoutIncMonRel` `\GTSoIMR`

This macro will draw a monochrome directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSoutIncidenceMonochromeRelation}{\mathrel{\GTSoIM/}}
\GTS@newcommand{\GTSoutIncMonoRel}{\mathrel{\GTSoIM/}}
\GTS@newcommand{\GTSoutIncMonRel}{\mathrel{\GTSoIM/}}
\GTS@newcommand{\GTSoIMR}{\mathrel{\GTSoIM/}}
```

`\GTSnegatedOutIncidenceMonochrome` `\GTSnegOutIncMono` `\GTSnegOutIncMon` `\GTSnOIM`

This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol.

```
\GTS@newcommand{\GTSnegatedOutIncidenceMonochrome}{%
\GTS@i{black}{black}{1}{1}{}%
}
\GTS@newcommand{\GTSnegOutIncMono}{\GTS@i{black}{black}{1}{1}{}}
\GTS@newcommand{\GTSnegOutIncMon}{\GTS@i{black}{black}{1}{1}{}}
\GTS@newcommand{\GTSnOIM}{\GTS@i{black}{black}{1}{1}{}}
```

`\GTSnegatedOutIncidenceMonochromeRelation` `\GTSnegOutIncMonoRel` `\GTSnegOutIncMonRel` `\GTSnOIMR`

This macro will draw a monochrome negated directed incidence upward (out(ward) incidence) symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSnegatedOutIncidenceMonochromeRelation}{%
\mathrel{\GTSnOIM/}%
}
\GTS@newcommand{\GTSnegOutIncMonoRel}{\mathrel{\GTSnOIM/}}
\GTS@newcommand{\GTSnegOutIncMonRel}{\mathrel{\GTSnOIM/}}
\GTS@newcommand{\GTSnOIMR}{\mathrel{\GTSnOIM/}}
```

`\GTSinIncidenceMonochrome` `\GTSinIncMono` `\GTSinIncMon` `\GTSiIM`

This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol. The operand on the left of the symbol should be considered the vertex to which the arc/the operand on the right of the symbol is incident.

```
\GTS@newcommand{\GTSinIncidenceMonochrome}{\GTS@i{black}{black}{}{}{1}}
\GTS@newcommand{\GTSinIncMono}{\GTS@i{black}{black}{}{}{1}}
\GTS@newcommand{\GTSinIncMon}{\GTS@i{black}{black}{}{}{1}}
\GTS@newcommand{\GTSiIM}{\GTS@i{black}{black}{}{}{1}}
```

\GTSinIncidenceMonochromeRelation \GTSinIncMonoRel \GTSinIncMonRel \GTSiIMR

This macro will draw a monochrome directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSinIncidenceMonochromeRelation}{\mathrel{\GTSiIM/}}
\GTS@newcommand{\GTSinIncMonoRel}{\mathrel{\GTSiIM/}}
\GTS@newcommand{\GTSinIncMonRel}{\mathrel{\GTSiIM/}}
\GTS@newcommand{\GTSiIMR}{\mathrel{\GTSiIM/}}
```

\GTSnegatedInIncidenceMonochrome \GTSnegInIncMono \GTSnegInIncMon \GTSnIIM

This macro will draw a monochrome negated directed incidence downward (in(ward) incidence) symbol.

```
\GTS@newcommand{\GTSnegatedInIncidenceMonochrome}{%
\GTS@i{black}{black}{1}{}{1}%
}
\GTS@newcommand{\GTSnegInIncMono}{\GTS@i{black}{black}{1}{}{1}}
\GTS@newcommand{\GTSnegInIncMon}{\GTS@i{black}{black}{1}{}{1}}
\GTS@newcommand{\GTSnIIM}{\GTS@i{black}{black}{1}{}{1}}
```

\GTSnegatedInIncidenceMonochromeRelation \GTSnegInIncMonoRel \GTSnegInIncMonRel \GTSnIIMR

This macro will draw a monochrome negated directed incidence downward (in(ward) incidence) symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSnegatedInIncidenceMonochromeRelation}{%
\mathrel{\GTSnIIM/}%
}
\GTS@newcommand{\GTSnegInIncMonoRel}{\mathrel{\GTSnIIM/}}
\GTS@newcommand{\GTSnegInIncMonRel}{\mathrel{\GTSnIIM/}}
\GTS@newcommand{\GTSnIIMR}{\mathrel{\GTSnIIM/}}
```

\GTSmetaNot \GTSmN

This macro will draw a meta-not or relation-not symbol.

```
\GTS@newprotectedcommand{\GTSmN}{%
\text{\ensuremath{%
\raisebox{-0.44ex}{\scalebox{1}[1]{%
\raisebox{0ex}{\scalebox{1}[1]{$\lnot$}}%
\hspace{-1.46ex}%
\raisebox{1.2ex}{\scalebox{0.7}[0.7]{m}}%
\hspace{0.2ex}%
}}%
}}%
}
\GTS@newcommand{\GTSmetaNot}{\GTSmN/}
```

\GTSmetaAnd \GTSmA

This macro will draw a meta-and or relations-and symbol. This is your duty to check that all relations have same arity. It should be easy as long as you remain in the realm of graphs/2-structures/binary structures. We didn't use Tarski's notation, since square cup has taken the preemptive meaning of disjoint union. Similarly, doublewedge or other wedge symbols have taken various meanings, whilst we want an unambiguous symbol.

```
\GTS@newprotectedcommand{\GTSmA}{%
\text{\ensuremath{%
\raisebox{-0.44ex}{\scalebox{0.95}[0.95]{%
\raisebox{0ex}{\scalebox{1}[1]{$\wedge$}}%
\hspace{-1.46ex}%
\raisebox{1.5ex}{\scalebox{0.7}[0.7]{m}}%
}}%
}}%
}
\GTS@newcommand{\GTSmetaAnd}{\GTSmA/}
```

\GTSmetaAndRelation \GTSmetaAndRel \GTSmAR

This macro will draw a meta-and symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSmetaAndRelation}{\mathrel{\GTSmA/}}
\GTS@newcommand{\GTSmetaAndRel}{\mathrel{\GTSmA/}}
\GTS@newcommand{\GTSmAR}{\mathrel{\GTSmA/}}
```

\GTSmetaOr \GTSmO

This macro will draw a meta-or or relations-or symbol. This is your duty to check that all relations have same arity. It should be easy as long as you remain in the realm of graphs/2-structures/binary structures ;). We didn't use Tarski's notation, since square cup has taken the preemptive meaning of disjoint union. Similarly, doublevee or other vee symbols have taken various meanings, whilst we want an unambiguous symbol.

```
\GTS@newprotectedcommand{\GTSmO}{%
\text{\ensuremath{%
\raisebox{-0.44ex}{\scalebox{0.95}[0.95]{%
\raisebox{0ex}{\scalebox{1}[1]{$\vee$}}%
\hspace{-1.46ex}%
\raisebox{1.5ex}{\scalebox{0.7}[0.7]{m}}%
}}%
}}%
}
\GTS@newcommand{\GTSmetaOr}{\GTSmO/}
```

\GTSmetaOrRelation
\GTSmetaOrRel
\GTSmOR

This macro will draw a meta-or symbol with relation spacing in math mode.

```
\GTS@newcommand{\GTSmetaOrRelation}{\mathrel{\GTSmO/}}
\GTS@newcommand{\GTSmetaOrRel}{\mathrel{\GTSmO/}}
\GTS@newcommand{\GTSmOR}{\mathrel{\GTSmO/}}
```

```
\endinput
⟨/package⟩
```

4 Acknowledgments

Thank You God! Thank You Father! Thank You Jesus! Thank You Holy-Spirit!

5 Support

This package is gratis and free. But if you can star it on GitHub, it is a nice encouragement.

6 Change History

Index